ALUForge'26 – 8-Bit ALU Hardware Hackathon

Digital Logic Based Arithmetic Processing System

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ABSTRACT

An Arithmetic Logic Unit (ALU) is a fundamental digital circuit used to perform arithmetic and logical operations on binary data. This project presents the implementation of an **8-bit ALU using basic digital logic ICs** on a breadboard.

The developed circuit accepts two 8-bit binary inputs and performs arithmetic operations based on the selected control input. The circuit uses logic gates, adders, registers and bus-interface components to process and transfer digital data.

The prototype was constructed and tested using discrete digital ICs. The operation of the circuit was verified by observing the output response and testing the generated signal using an oscilloscope.

This project provides practical understanding of **binary arithmetic, logic gates, digital ICs, data transfer, control signals and hardware-level implementation of an ALU**.

OBJECTIVES

Hardware Design

- To design an 8-bit ALU using standard digital logic ICs.
- To understand the hardware implementation of binary arithmetic operations.

- To construct and test the circuit using a breadboard-based prototype.

Arithmetic Operations

- To perform 8-bit addition.
- To implement subtraction using 2's complement arithmetic.
- To understand carry propagation between 4-bit adder stages.

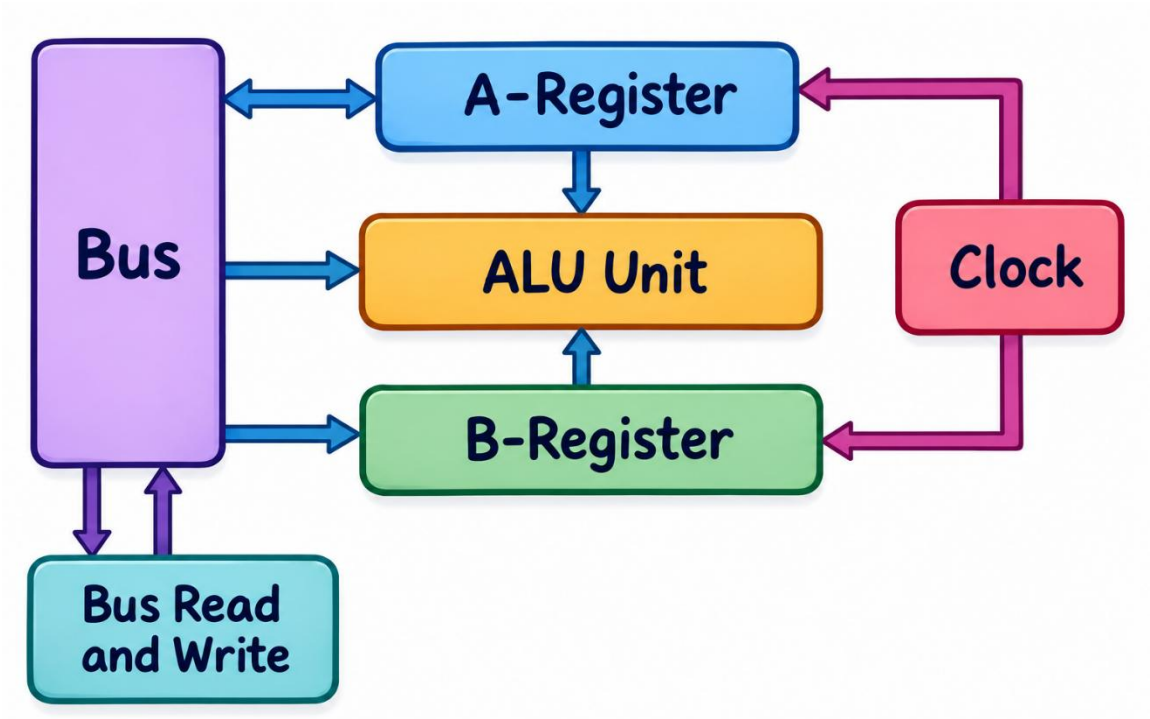
Data Handling

- To transfer 8-bit data through a common bus.
- To use registers for storing required data and status information.
- To control the direction and availability of data using appropriate control signals.

Practical Verification

- To observe the output of the ALU for different input combinations.
- To verify Carry Flag and Zero Flag conditions.
- To test the circuit using actual hardware measurements.

BLOCK DIAGRAM



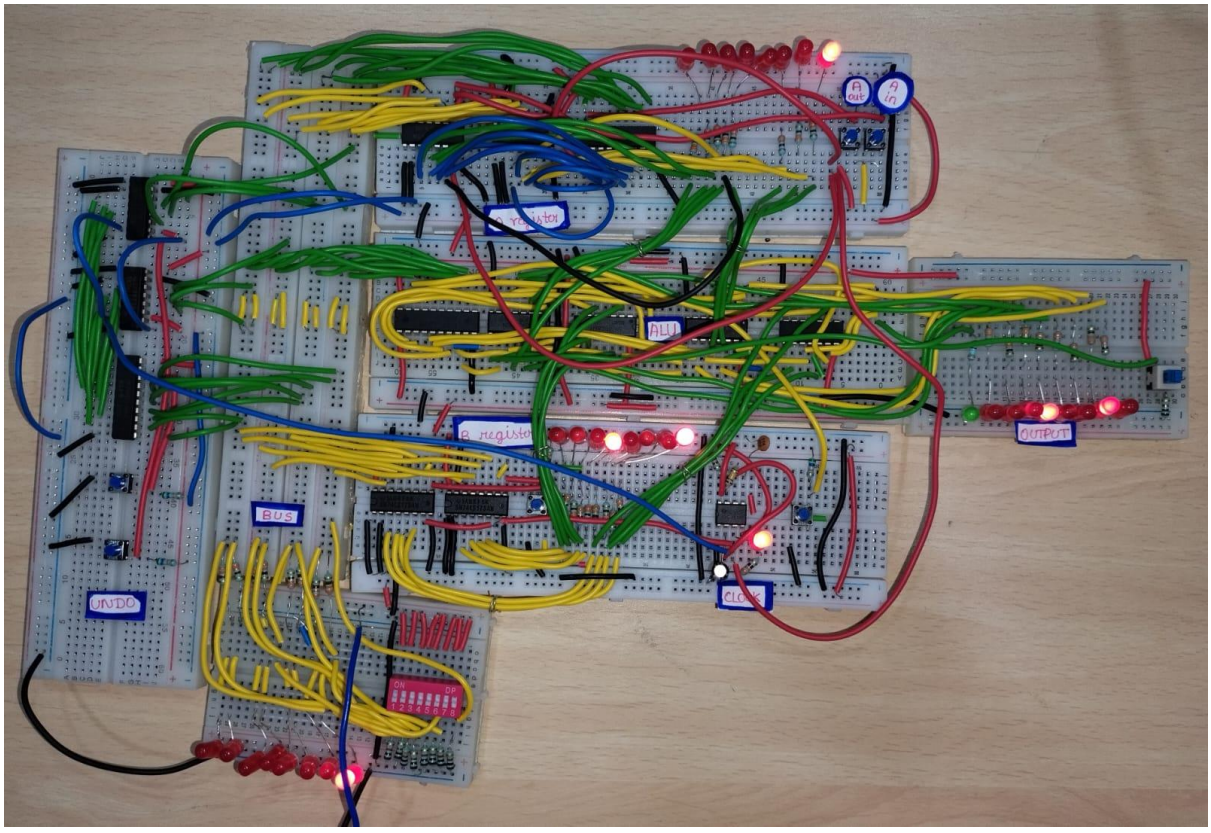
PROJECT OVERVIEW

The developed ALU is organized as a combination of input storage, arithmetic processing, data transfer and control sections. Two 8-bit operands are supplied to the circuit and are held in the required register sections before processing.

The B operand passes through XOR gates whose operation is controlled by the **SU signal**. In addition mode, the B value is passed without modification. In subtraction mode, the B value is complemented and an additional carry-in is used to implement 2's complement subtraction.

Two 4-bit adder ICs are cascaded to perform the complete 8-bit arithmetic operation. The resulting data is transferred through the bus interface. The circuit also provides status information such as Carry Flag and Zero Flag.

IMPLEMENTED CIRCUIT



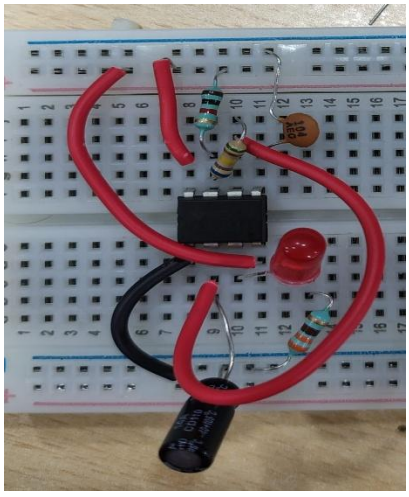
The proposed ALU was assembled using discrete digital logic ICs on a breadboard. The circuit combines arithmetic, logic, register and bus-interface sections to form an 8-bit data processing unit. Separate ICs are used for XOR operations, binary addition, bus transfer and flag storage.

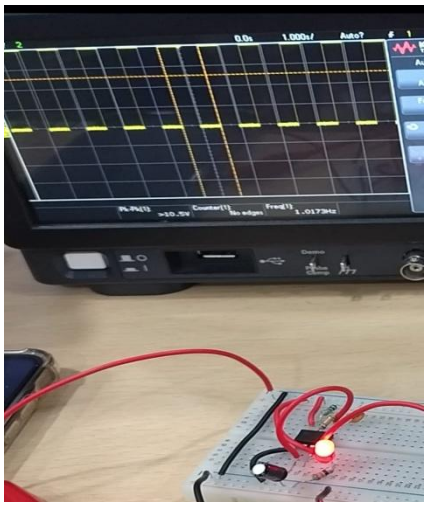
IC	Quantity	Purpose
IC NE555P	1	Timer/clock generation, approximately 1 Hz in your clock section
IC SN74HC86N	2	XOR gates; control/invert the B input for ADD/SUB operation
IC SN74LS283N	2	4-bit binary adders; two ICs together perform 8-bit addition/subtraction
IC SN74HCT245	2	8-bit bidirectional bus transceiver; transfers data between ALU/register side and BUS
IC SN74LS173AN	4	4-bit register used as the **Flags Register**

CLOCK

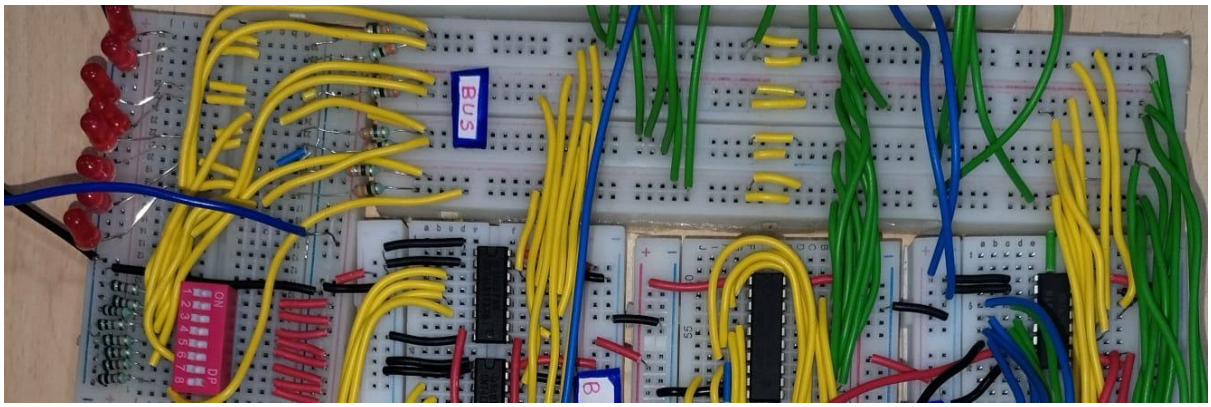
The 555 timer is used to generate the clock signal for the sequential sections of the circuit.

The clock provides a regular timing reference so that register operations occur at controlled instants.





COMMON 8-BIT BUS



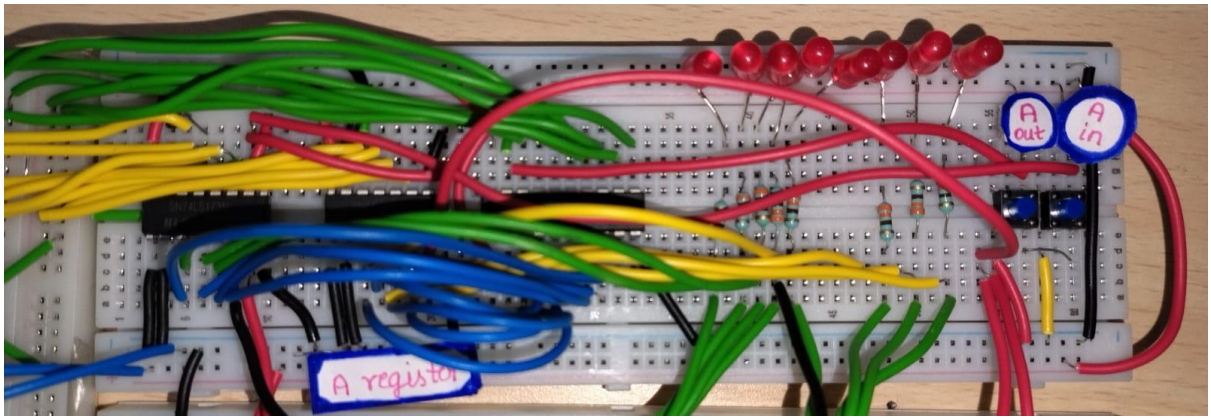
The common bus is an 8-bit group of parallel signal lines used to transfer binary data between different sections of the circuit. Instead of using separate connections for every data-transfer path, the bus provides a shared communication path**.**

A-REGISTER

The A register is used to hold the first operand required by the arithmetic section. Storing the operand separately prevents changes at the input from directly affecting the value being processed.

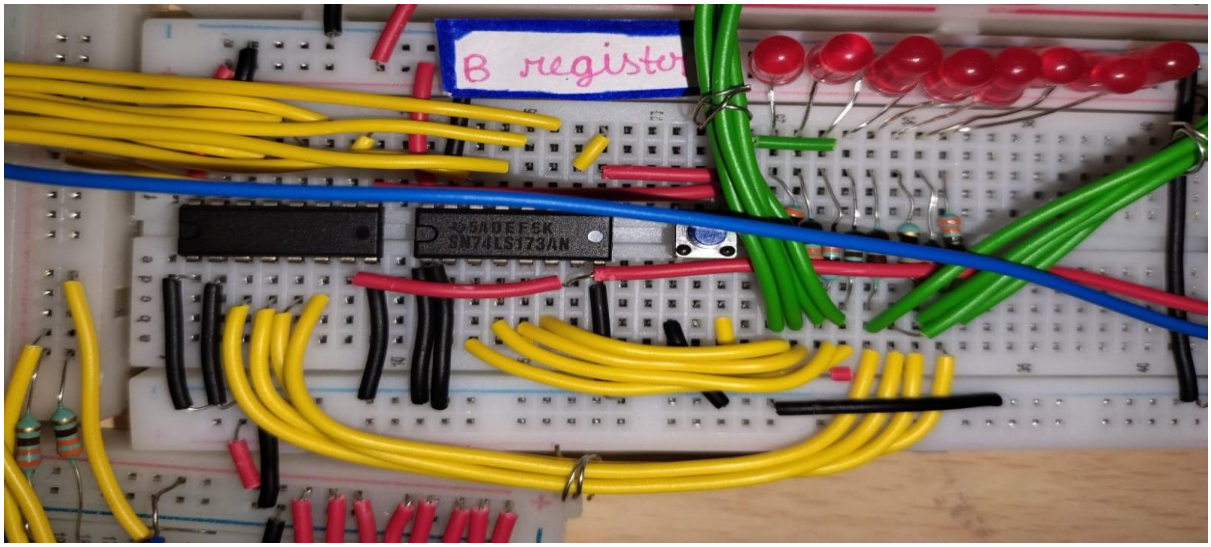
Once the required value is loaded, the register maintains that value until it is replaced by another input. The stored A operand is then supplied to the ALU during the selected operation.

Function : Stores and provides the first 8-bit operand for the ALU operation.



B-REGISTER

The B register holds the second operand used by the ALU. Keeping A and B as separate operands allows the arithmetic section to process them simultaneously. The B value is subsequently passed through the XOR section when selecting between addition and subtraction. Function : Stores and provides the second 8-bit operand for arithmetic processing.



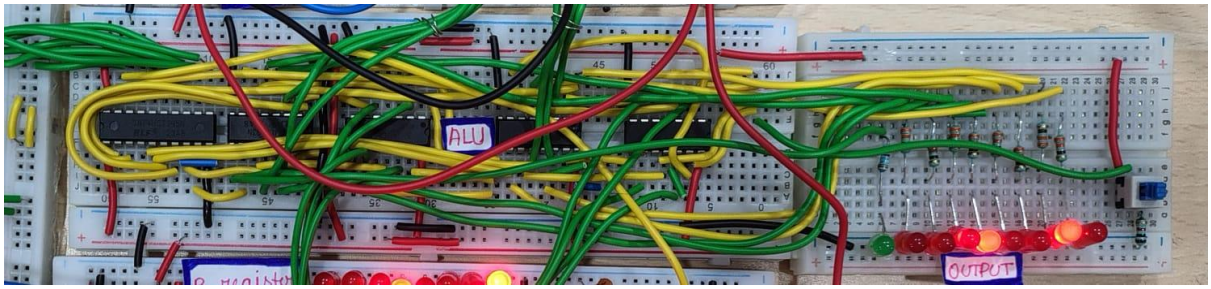
ALU UNIT

The ALU is the main processing section of the project.

It receives the A and B operands and produces the required 8-bit arithmetic result.

The arithmetic section uses:

- 74LS86 → XOR operation
- 74LS283 × 2 → 8-bit addition
- Control signal SU → operation selection



WORKING PROCEDURE

Step 1 – Provide the operands

The two required binary operands are supplied to the input sections of the circuit.

For example:

A = 21 = 00010101

B = 54 = 00110110

Step 2 – Store the operands

The required operand values are held in the register sections so that they remain stable during the arithmetic operation.

Step 3 – Select the operation

The SU control signal determines whether the circuit performs addition or subtraction.

SU = 0 → Addition

SU = 1 → Subtraction

Step 4 – Process B operand

The B operand passes through the XOR gates.

For addition:

B XOR 0 = B

For subtraction:

B XOR 1 = NOT B

Step 5 – Perform arithmetic

The two 74LS283 ICs process the lower and upper four bits to obtain the complete 8-bit result.

Step 6 – Transfer the result

The result is made available through the bus interface using the 74LS245 transceiver.

Step 7 – Observe status

The resulting operation is used to determine status information such as:

- Carry Flag
- Zero Flag

RESULT

- The implemented hardware successfully demonstrates the basic operation of an 8-bit ALU using discrete digital ICs.
- The circuit accepts two binary operands, performs the selected arithmetic operation, transfers the result through the bus and provides status information through the flag section.
- Hardware testing was carried out by observing the clock signal, data movement and arithmetic output.

CONCLUSION

- This project provided a practical understanding of how an 8-bit ALU can be constructed using basic digital logic ICs.
- The implementation demonstrated the interaction between registers, XOR gates, binary adders, bus transceivers, flag storage and clock generation.
- The hardware prototype helped verify the flow of binary data and arithmetic operations at the circuit level.
- Through this implementation, the project demonstrates the fundamental principles involved in the data path and control sections of a simple digital processing system.